

Samuel Babichenko

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Education

PhD in Statistics and Applied Probability, UC Santa Barbara · Sep 2021 – Jun 2026

Advisor: Tomoyuki Ichiba.

Teaching assistant for PSTAT 176/276. Mentored 10+ undergraduate researchers.

BS in Mathematics, UC San Diego · Sep 2019 – Jun 2021

Dean's Undergraduate Excellence Award (top 0.5%, Physical Sciences).

Research

Noise-State Calculus for Dynamic Games with Strategic Information · PhD dissertation, 2026

· sbabichenko.com/dissertation

Continuous-time linear-quadratic-Gaussian games in which players learn from each other's actions. Each player keeps estimates of the primitive shocks instead of belief hierarchies, so beliefs, prices and policies are impulse responses and equilibrium is a fixed point in them, checked against arbitrary deviations. The information wedge, the shadow price of changing an opponent's beliefs, explains where these games depart from their full-information versions. Applications to Kyle–Back markets with several informed traders, networks of local markets, delayed public signals, and deviations that only some players can detect.

Forecasting and Manipulating the Forecasts of Others · arXiv: 2603.12140

Solo-authored, submitted March 2026. The two-player core of the dissertation: an exact solution, with no large-population limit, to a class of games with private signals open since Townsend (1983). Splits the cost of dispersed information into an estimation part and a strategic part, and finds cases where more precise private information raises total cost.

Software. The noisestate solver in C++, about 10 ms per equilibrium, also compiled to WebAssembly for the browser explorer at sbabichenko.com/noisestate. Python package: pip install noisestate.

Talk. UCSD Stochastic Systems Seminar.

Experience

Wells Fargo, Charlotte, NC · *Quantitative Researcher, rotational program* · Jul 2026 – present

- Unifying the team's backtesting workflow.
- Refining the prepayment trading signal from my internship with the gated Decision Mesh.

Wells Fargo, Charlotte, NC · *Quantitative Researcher Intern, Mortgage Model Development* · Jun – Aug 2025

- Delivered the assigned mortgage pool segmentation early, then built a tree-based alternative with diagnostic tools. The buy-side trading desk then brought me onto an open-ended problem.
- Found that the team's prepayment models used normal approximations that miss the binomial structure of loan-level prepayment. Built a binomial model that corrects for effective loan count and partitioned pools to isolate systematic errors. The errors were persistent ($R^2 \approx 0.8$ at a 5-month horizon), which makes them usable as a trading signal.

Selected projects

Decision Mesh · sbabichenko.com/gate · github.com/sbabichenko/Decision-Mesh

Mesh regression that fits continuous surfaces where trees fit steps. A right-triangle or rectangular mesh is refined only where a false-discovery gate says the data justify a cut, and stops on its own. Built for the discontinuities that tree models showed in the prepayment work.

Multi-agent traffic simulation (Waymo Open Motion Dataset)

Reimplemented BehaviorGPT, a decoder-only transformer with agent-agent attention, for closed-loop 10 Hz trajectory generation. Extending it so the transformer predicts only at branch points where behavior becomes multimodal and a conditional diffusion model fills in between.

Skills

Programming: C++23, Python, CUDA, SQL, WebAssembly, Linux

ML and data: PyTorch, transformers, diffusion models, LightGBM, XGBoost, Polars, NumPy

Methods: stochastic calculus and control, Kalman filtering, Monte Carlo, PDE methods, hierarchical Bayesian models, time series, derivatives pricing